



Vertex-edge domination in unit disk graphs[☆]

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ABSTRACT

Let $G = (V, E)$ be a simple undirected graph. A set $D \subseteq V$ is called a vertex-edge dominating set of G if for each edge $e = uv \in E$, either u or v is in D or one vertex from their neighbor is in D . Simply, a vertex $v \in V$, vertex-edge dominates every edge uv , as well as every edge adjacent to these edges. The objective of the vertex-edge domination problem is to find a minimum size vertex-edge dominating set of G . Herein, we study the vertex-edge domination problem in unit disk graphs and prove that the decision version of the problem belongs to the NP-complete class. We also show that the problem admits a simple polynomial-time 4-factor approximation algorithm and a polynomial-time approximation scheme (PTAS) in unit disk graphs.

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1. Introduction

Let $G = (V, E)$ be a simple undirected graph. The *open neighborhood* of a vertex $v \in V$ in G is the set $N_G(v) = \{u \in V \mid uv \in E\}$ whereas the *closed neighborhood* is the set $N_G[v] = N_G(v) \cup \{v\}$. A *dominating set* D of G is a subset of V such that every vertex in V is in D or adjacent to at least one vertex in D . A vertex $v \in D$ *dominates* all its neighbors and itself. The goal of the *domination problem* is to find a minimum cardinality subset $D \subseteq V$ such that D dominates all the vertices of G .

An edge $uv \in E$ is said to be vertex-edge dominated by a vertex $w \in V$ if w is in closed neighborhood of either u or v i.e., $w \in N_G[u] \cup N_G[v]$. A *vertex-edge dominating set* (VEDs) of a simple undirected graph $G = (V, E)$ is a set $D \subseteq V$ of G such that every edge $e \in E$ is vertex-edge dominated by at least one vertex in D . The *VEDs problem* asks to find a VEDs of minimum size in a given graph. A set $D \subseteq V$ is a *double vertex-edge dominating set* if every edge $e \in E$ is vertex-edge dominated by at least two vertices in D . A set $D \subseteq V$ is called a *total vertex-edge dominating set* if every edge $e \in E$ is vertex-edge dominated by D and the graph induced by D has no isolated vertices.

One of the real-time applications of the VEDs problem is as follows: consider a client server architecture based model in a computer networking system. A client can communicate with a server either directly or through some intermediate clients and servers and vice versa. In a secure-based system, where security and privacy of data communication are utmost needed, for example, military agencies data communication, the secure transmission of data is the highest priority. In that case, to secure the communication of the whole networking system, one has to secure each and every communication line between any two adjacent pairs of clients and servers. We refer a communication line between any two adjacent pairs of clients or servers is secure if and only if one of them is a secure agent or any one of its open neighborhood is a secure agent. A secure agent can secure the communication line not only with the clients or servers who are directly linked to them but also with the clients or servers who are connected at a two-hop distance from them. To secure the whole transmission system one can deploy secure agents to all clients or servers attached to the system. As the deployment

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of secure agents is too costly we may need to minimize the deployment of secure agents in the transmission system. The smallest group of secure agents with this property is a minimum vertex-edge dominating set for the graph which represents the computer network.

1.1. Related work

The vertex-edge dominating set problem was introduced by Peters [20] and then studied further by different researchers. In particular, bounds on the vertex-edge domination number in several graph classes were studied in [3,16–18,22], vertex-edge degrees and vertex-edge domination polynomials of different graphs were discussed in [5,11,25,26], whereas the relations between some vertex-edge domination parameters were discussed in [3,6,14,17,18], several algorithmic aspects were discussed in [17]. Some variants of vertex-edge domination problem were studied in [2,7,13,15,21].

The minimum cardinality of a vertex-edge dominating set (double vertex-edge dominating set, respectively) of G is termed the *vertex-edge domination number* and denoted by $\gamma_{ve}(G)$ (the *double vertex-edge domination number*, $\gamma_{dve}(G)$, respectively). Krishnakumari et al. [16] proved that for every tree T of order $n \geq 3$ with ℓ leaves and s support vertices, we have $\frac{(n-\ell-s+3)}{4} \leq \gamma_{ve}(T) \leq \frac{n}{3}$. In [13], Krishnakumari et al. showed that determining $\gamma_{dve}(G)$ for bipartite graphs is NP-hard, whereas for every non-trivial connected graphs G , $\gamma_{dve}(G) \geq \gamma_{ve}(G) + 1$, and for every tree T , we have $\gamma_{dve}(T) = \gamma_{ve}(T) + 2$. They also provided two lower bounds on the double vertex-edge domination number of trees and unicycle graphs in terms of order n , the number of leaves and support vertices, respectively.

Boutrig et al. [3] presented a new relationship between the vertex-edge domination and some other domination parameters, answering the four open questions posed by Lewis [17]. Then, for every non-trivial connected $K_{1,k}$ -free graph, with $k \geq 3$, they provided an upper bound for the independent vertex-edge domination number in terms of the vertex-edge domination number and showed that for every non-trivial tree the independent vertex-edge domination number can be bounded by the domination number. For connected C_5 -free graphs, they also established an upper bound on the vertex-edge domination number. Next Boutrig and Chellali [2] studied the total vertex-edge domination. The minimum cardinality of a total vertex-edge dominating set of graph G called the *total vertex-edge domination number* and denoted by $\gamma_{ve}^t(G)$. They showed that determining $\gamma_{ve}^t(G)$ for bipartite graphs is NP-hard, and in case of tree T different from a star having order n , with ℓ leaves and s support vertices, respectively, we have $\gamma_{ve}^t(G^T) \leq \frac{(n-\ell+s)}{2}$. In the same article, they established a necessary condition for a graph G to satisfy $\gamma_{ve}^t(G) = 2\gamma_{ve}(G)$ and for a tree T , $\gamma_{ve}^t(T) = 2\gamma_{ve}(T)$.

Later Venkatakrishnan and Kumar [24] proved that the minimum double vertex-edge dominating set problem is NP-hard for chordal graphs and APX-hard for bipartite graphs with maximum degree 5. They also proposed a linear-time algorithm for finding a minimum double vertex-edge dominating set in proper interval graphs. In addition, they showed that the minimum double vertex-edge dominating set problem cannot be approximated the factor $(1 - \epsilon) \ln |V|$ for any $\epsilon \geq 0$ unless $NP \subseteq DTIME(|V|^{O(\ln \ln |V|)})$. Finally, the influence of edge removal, edge addition, and edge subdivision on the double vertex-edge domination number of a graph was investigated by Krishnakumari and Venkatakrishnan [14]. Next, Horoldagva et al. [11] obtained some results on the regularity and irregularity of vertex-edge and edge-vertex degrees in graphs. Recently, Żyliński [27] proved that for any connected graph G of order $n \geq 6$, $\gamma_{ve}(G) \leq \lfloor \frac{n}{3} \rfloor$.

1.2. Our contribution

We study the VEDs problem in unit disk graphs. A *unit disk graph* (UDG) is the intersection graph of equal-radii disks in the plane. Given a set $S = \{d_1, d_2, \dots, d_n\}$ of n circular disks in the plane, each having diameter 1, the corresponding UDG $G = (V, E)$ is defined as follows: each vertex $v_i \in V$ corresponds to the disk $d_i \in S$, and there is an edge between two vertices if and only if the relevant pair of disks intersect.

We show that the decision version of the VEDs problem is NP-complete in unit disk graphs (Section 2). We propose a simple polynomial-time 4-factor approximation algorithm (Section 3), and a polynomial-time approximation scheme for the VEDs problem (Section 4). Finally, we conclude the paper in Section 5.

2. NP-hardness

In this section, we show a polynomial-time reduction from an arbitrary instance of the *vertex cover* problem in planar graphs with degree at most 3 which is known to be NP-hard [9] to an instance of the VEDs problem to prove that the later one is also NP-hard. The decision versions of both these problems are defined below.

The VEDs problem on UDGs (VEDS-UDG)

Instance: A unit disk graph $G = (V, E)$ and a positive integer k .

Question: Does there exist a vertex-edge dominating set D of G such that $|D| \leq k$?

The vertex cover problem on planar graphs (VC-PLA)

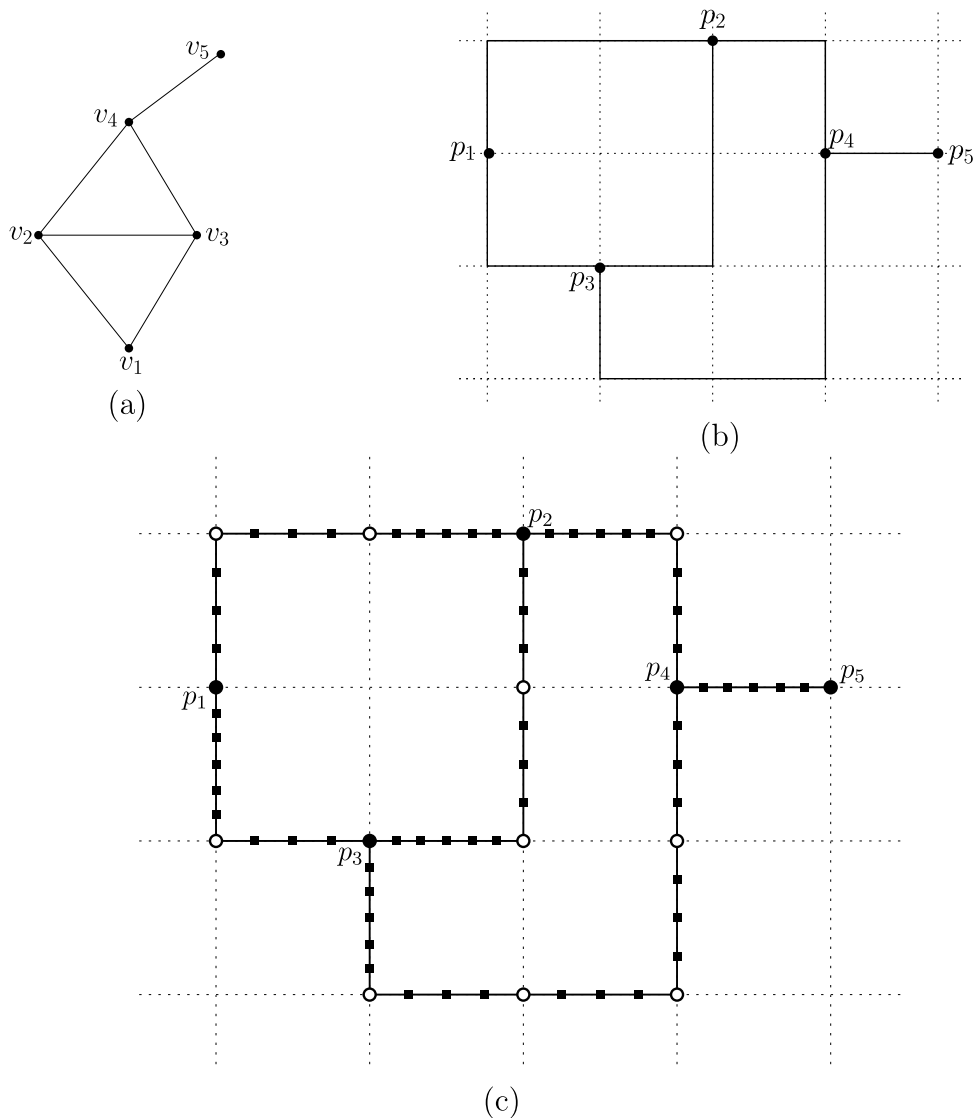


Fig. 1. (a) A planar graph G of degree 3, (b) its embedding on a grid, and (c) a UDG construction from the embedding.

Instance: A planar graph G having maximum degree 3 and a positive integer k .

Question: Does there exist a vertex cover C of G such that $|C| \leq k$?

Lemma 2.1 ([23]). A planar graph $G = (V, E)$ with maximum degree 4 can be embedded in the plane using $O(|V|^2)$ area in such a way that its vertices are at integer coordinates and its edges are drawn so that they are along the grid line segments of the form $x = i$ or $y = j$, for some i and j .

This embedding is known as the *orthogonal drawing* of a graph. There is a linear-time algorithm given by Biedl and Kant [1] that gives an orthogonal drawing of a given graph with at most 2 bends along each edge (see Fig. 1).

Corollary 2.2. A planar graph $G = (V, E)$ with maximum degree 3 and $|V| \geq 3$ can be embedded in the plane with its vertices are at $(4i, 4j)$ and its edges are drawn as a sequence of consecutive axis-parallel line segments on the lines $x = 4i$ or $y = 4j$, for some i and j .

Lemma 2.3. Let $G = (V, E)$ be an instance of VC-PLA with $|E| \geq 2$. An instance $G' = (V', E')$ of VEDS-UDG can be constructed from G in polynomial-time.

Proof. Our construction of G' from G is done in three steps.

Step 1: Embedding graph G into a grid of size 4×4 . G can be embedded in the plane using one of the algorithms in [10,12] (see Lemma 2.1 and Corollary 2.2) with each of its edges as a sequence of connected line segment(s) of length four units. Let ℓ be the total number of line segments used in the embedding. The points $\{p_1, p_2, \dots, p_n\}$ are termed *node points* in the embedding correspond to the vertex set $V = \{v_1, v_2, \dots, v_n\}$ (see Fig. 1(a) and 1(b)).

Step 2: Adding extra points. For each edge $p_i p_j$ having length 4 units, (i) we add two points α and β on the edge $p_i p_j$ such that α is 0.8 unit apart from p_i and β is 0.8 unit apart from p_j , and (ii) add another three points between α and β with distance 0.6 unit from each other, respectively (thus adding five points in total, see edge $p_4 p_5$ in Fig. 1(c)). For each edge of length greater than 4 units, we also add points as follows: (i) add a point in the joining point (grid point) of each line segments other than the *node points* and name it as a *joint point* (see empty circular points in Fig. 1(c)), (ii) for one of the two line segments whose one endpoint is associated with node point, we add five points at distances 0.8, 1.4, 2, 2.6 and 3.2 units from the node point, and for other line segments, we add three points at distance 1 units from each other excluding the *joint points* (see edge $p_3 p_4$ in Fig. 1(c)). We name the points added in this step as *added points*.

Step 3: Construction step. For convenience, denote the set of node points by N and set of added points by A , respectively, that is, $N = \{p_i \mid v_i \in V\}$ and $A = \{q_1, q_2, \dots, q_{4\ell+|E|}\}$. We construct a UDG $G' = (V', E')$, where $V' = N \cup A$ and there is an edge between two points in V' if and only if the Euclidean distance between the points is at most 1 (see Fig. 1(c)). Observe that $|N| = |V| (= n)$ and $|A| = 4\ell + |E|$, where ℓ is the total number of line segments in the embedding and $|E|$ is the total number of edges in G . Since G is planar, $|E| = O(n)$. It also follows from Lemma 2.1 that $\ell = O(n^2)$. Therefore both $|V'|$ and $|E'|$ are bounded by $O(n^2)$, and hence G' can be constructed in polynomial-time. $\square$

Theorem 2.4. VEDS-UDG is NP-complete.

Proof. For any given set $D \subseteq V$ and a positive integer k , we can verify in polynomial-time whether D is a vertex-edge dominating set of size at most k by checking whether each edge in E is vertex-edge dominated by a vertex in D or not. Hence, VEDS-UDG $\in$ NP.

Now, we need to prove VEDS-UDG $\in$ NP-hard. For the hardness proof, we show a polynomial-time reduction from VC-PLA to VEDS-UDG. Let $G = (V, E)$ be an instance of VC-PLA. Construct the instance $G' = (V', E')$ of VEDS-UDG as discussed in Lemma 2.3. We have the following claim.

Claim. G has a vertex cover of size at most k if and only if G' has a vertex-edge dominating set of size at most $k + \ell$.

Necessity. Let $C \subseteq V$ be a vertex cover of G such that $|C| \leq k$. Let $N' = \{p_i \in N \mid v_i \in C\}$, i.e., N' is the set of vertices in V' that correspond to the vertices in C . The idea is to choose one vertex from each segment in the embedding such that the chosen vertex set $A' (\subseteq A)$ together with N' , i.e., $N' \cup A'$ will form a VEDs of cardinality $k + \ell$ in G' . As C is a vertex cover in G , every edge in G has at least one of its endpoints in C . Let $v_i v_j \in E$ and assume $v_i \in C$ (the same argument works for $v_j \in C$ or if both v_i and $v_j \in C$). It follows from the construction of G' that the edge $p_i p_j$ is represented as a sequence of line segments in the graph G' , where p_i and p_j are nodes in G' corresponding to vertices v_i and v_j in G . Start traversing the segments from p_i , and add each fourth vertex to A' encountered from p_i to p_j in the traversal (see Fig. 2 for an illustration, where both big circles and squares belong to A' while traversing from p_1 to p_2 , p_2 to p_3 , p_1 to p_3 , p_4 to p_2 , p_4 to p_5 and p_4 to p_3 , respectively).

Apply the same process to each chain of line segments in G' corresponding to each edge in G . Observe that the cardinality of A' is ℓ as we have chosen one vertex from each segment in the embedding. Let $D = N' \cup A'$. Now, observe that D is a vertex-edge dominating set in G' as each edge in G' is vertex-edge dominated by at least one vertex in D and $|D| = |N'| + |A'| \leq k + \ell$ as required.

Sufficiency. Let $D \subseteq V'$ be a VEDs of size at most $k + \ell$. We argue that G has a vertex cover of size at most k based upon the following

Claim: (i) at least one vertex on each segment in the embedding must belong to D and hence $|A \cap D| \geq \ell$, where ℓ is the total number of segments in the embedding. We shall show that, by removing and/or replacing some vertices in D , a set of at most k vertices from N can be chosen such that the corresponding vertices in V is a vertex cover of G . Let $C = \{v_i \in V \mid p_i \in D \cap N\}$. If any edge $v_i v_j$ in G has none of its end vertices in C , then consider the points p_i and p_j corresponding to v_i and v_j respectively.

Case (i): If p_i is the only vertex that is connected with p_j in G' , then the chain of segments (say ℓ') in the path $p_i \rightsquigarrow p_j$ in G' has at least $\ell' + 1$ vertices in D (see Fig. 3(a) for example). In this case, we delete one point from the segment containing two points in D and introduce p_i in D .

Case (ii): If both p_i and p_j are connected with some points p_k and p_ℓ respectively in G' , then either the chain of segments (say ℓ') in the path $p_i \rightsquigarrow p_j$ in G' has at least $\ell' + 1$ vertices in D (similar to Case (i)) or the chain of segments (say ℓ') in both the path $p_i \rightsquigarrow p_k$ and $p_j \rightsquigarrow p_\ell$ in G' has at least $\ell' + 1$ vertices in D (see Fig. 3(b) for example). In this case, **Case (a):** If p_i is the only vertex that is connected with p_k , such that $p_k \notin D$ (similar case applies if p_j is the only vertex connected with p_ℓ and $p_\ell \notin D$), then we choose the segment from $p_i \rightsquigarrow p_k$ having two vertices in D and remove one vertex of the segment from D and introduce p_i in D . **Case (b):** If $p_k \in D$ (similar case applies if $p_\ell \in D$), then remove one vertex from the segment having its two vertices in D in the path $p_i \rightsquigarrow p_k$ and introduce p_j in D . **Case (c):** If Case (a) and Case (b) fail, then arbitrarily choose p_i or p_j in D by removing one vertex from the segment containing its two vertices in D from the path $p_i \rightsquigarrow p_k$ or $p_j \rightsquigarrow p_\ell$, respectively. Update C and repeat the process till every edge has at least one of its end vertices in C . Due to Claim (i), C is a vertex cover in G with $|C| \leq k$. Therefore, VEDS-UDG is NP-hard.

As VEDS-UDG is in NP as well as NP-hard, VEDS-UDG is NP-complete. $\square$

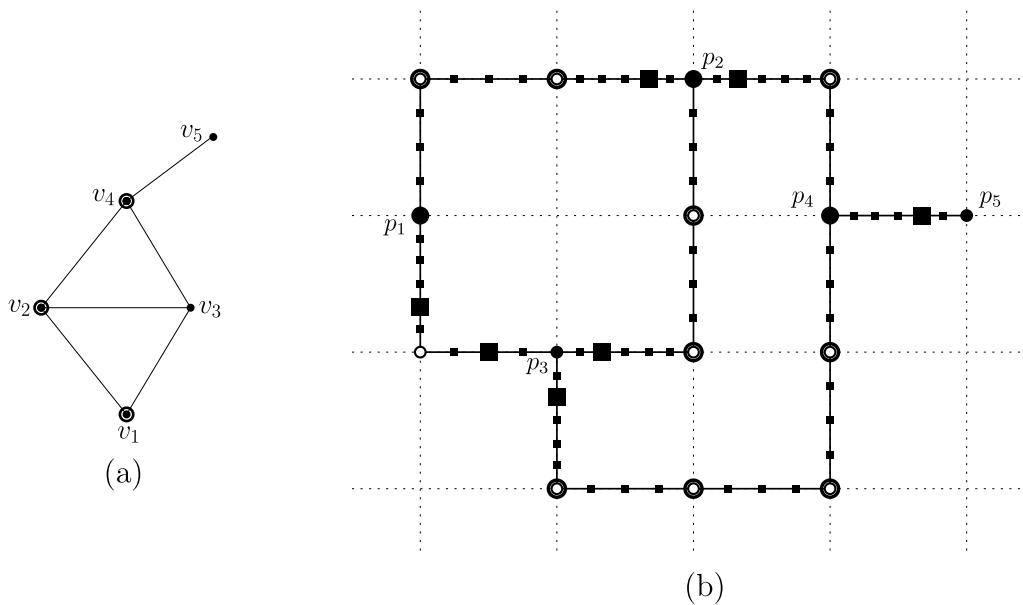


Fig. 2. (a) A vertex cover $\{v_1, v_2, v_4\}$ of G , and (b) the construction of A' in G' .

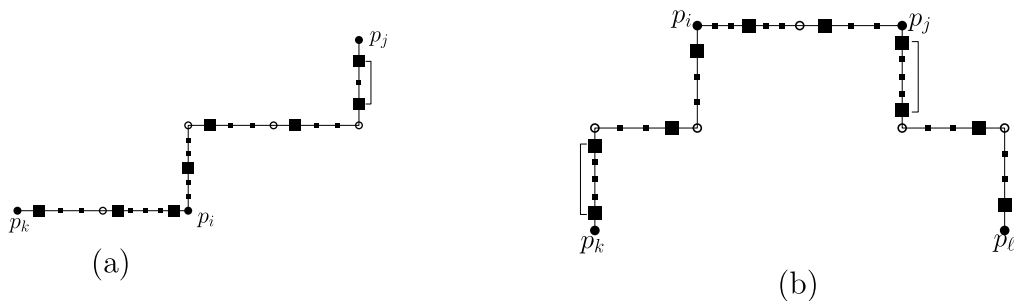


Fig. 3. (a) p_i is connected with only p_j , (b) p_i is connected with p_k and p_j is connected with p_l .

3. Approximation algorithm

Let $P = \{p_1, p_2, \dots, p_n\}$ (center of UDG G) be the point set given as the input and $\mathcal{R}$ be an axis-parallel rectangular region containing the point set P . We add an edge between two points if and only if the distance between two points is at most 1. We present a 4-factor approximation algorithm for the VEDs problem in unit disk graphs with the help of the 4-color partitioning technique given by De et al. [8]. Carmi et al. [4] used a similar technique to obtain a 4-factor approximation algorithm for minimum dominating set problem in unit disk graphs, where they used the hexagonal partitioning of $\mathcal{R}$ instead of the square partitioning given by De et al. to improve the time complexity.

Here, we partition $\mathcal{R}$ into regular hexagonal cells, where the side length of each regular hexagonal cell is $\frac{1}{2}$ (see Fig. 4(a)). After partitioning $\mathcal{R}$, we use the 4-color partitioning technique to propose a 4-factor approximation algorithm for the VEDs problem in unit disk graphs. The proposed algorithm runs in polynomial-time.

Definition 3.1 (Edge of a Cell). By an edge of a cell, we mean an edge of G with at least one end point is within that cell (for example, see all the edges shown in Fig. 4(b), these edges are the edge of the cell containing p_i).

Claim 3.2. Any point in a single cell can vertex-edge dominate all the edges of that cell.

Proof. Since the distance between any pair of points inside a cell is at most 1, each point in a cell can vertex-edge dominate all the edges of that cell (see Fig. 4(b), p_i vertex-edge dominates all the edges of the cell containing p_i). $\square$

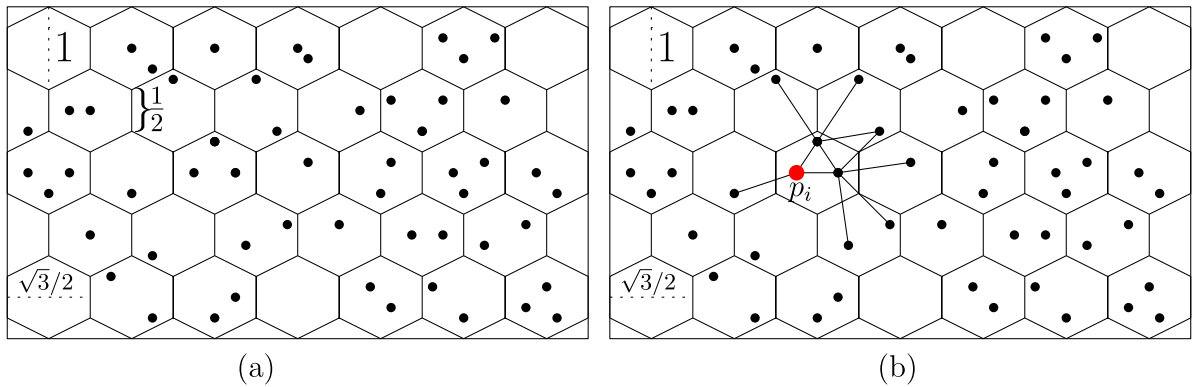


Fig. 4. (a) $\mathcal{R}$ containing the point set P partitioned into regular hexagons, and (b) a point p_i vertex-edge dominate the edges in the cell containing p_i .

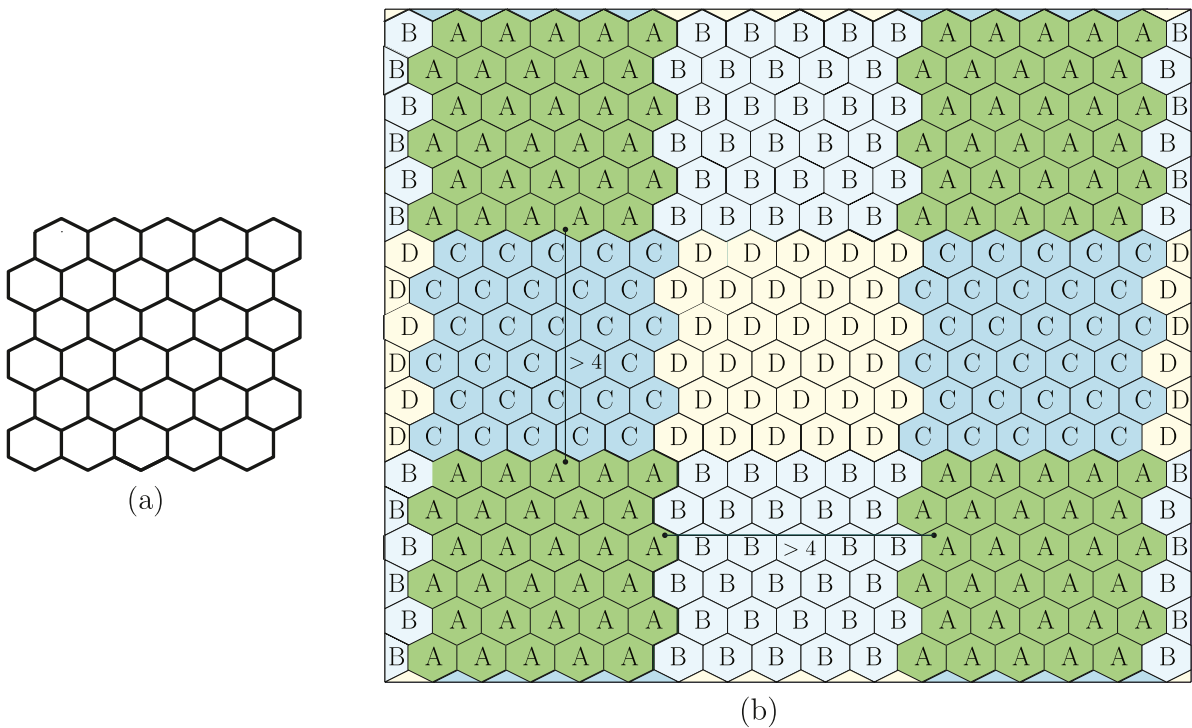


Fig. 5. (a) A combined-cell, and (b) a combined-cell partition of $\mathcal{R}$ and its 4-coloring scheme.

Definition 3.3 (Combined-cell). A combined-cell can be defined as a combination of 6 consecutive rows of regular hexagonal cells, where each row consists of 5 consecutive regular hexagons, resulting 30 hexagonal cells in a combined-cell (see Fig. 5(a) for example).

The idea behind our algorithm is as follows: consider a combined-cell partitioning of $\mathcal{R}$ such that no point of P lies on the boundary of any combined-cell, and a 4-color partitioning of it (see Fig. 5(b)). The 4-color combined-cell partitioning of $\mathcal{R}$ color all the combined-cells such that if a combined-cell S is assigned the color D then its adjacent combined-cells are assigned different three colors, say A, B, and C, such that a pair of opposite combined-cells adjacent to S are assigned the same color (see Fig. 5(b)).

Claim 3.4. If S' and S'' are any two combined-cells of the same color, then the minimum distance between any two points $p \in S' \cap P$ and $q \in S'' \cap P$ is greater than 4.

Proof. Follows from the fact that the size of a combined-cell and no point of P lies on the boundary of any combined-cell (see Fig. 5(b)). $\square$

Claim 3.4 leads to the fact that there does not exist a single point which can simultaneously vertex-edge dominate an edge connecting a point from S' and another edge connecting a point from S'' . Therefore, we can say that if OPT_A denotes a minimum size subset of points in P such that OPT_A is a VEDs of all the combined-cells colored with A , then OPT_A is the union of the optimum solutions of the VEDs problems for all the combined-cells colored with A . The same result holds for all other three colors also. Now we consider each combined-cell S and find the minimum size subset of P , which ensures the VEDs of S . Finally, we report the set T which is the union of the solutions for all the combined-cells. The pseudocode of the algorithm is described in Algorithm 1.

Algorithm 1 Vertex_Edge_Domination(P)

Require: A point set P

Ensure: A vertex-edge dominating set T for the UDG corresponding to P .

- 1: Let $\mathcal{R}$ be the minimum size axis-parallel rectangle that contains all the points in P .
 - 2: $T \leftarrow \emptyset$
 - 3: Partition $\mathcal{R}$ into combined-cells defined in Definition 3.3
 - 4: **FOR** each combined-cell $S \in \mathcal{R}$ with $P \cap S \neq \emptyset$
 - 5: $S_0 \leftarrow \text{Vertex_Edge_Domination_Single_Combined-cell}(P, S)$ ▷ call Algorithm 2
 - 6: $T \leftarrow T \cup S_0$
 - 7: **ENDFOR**
 - 8: Return T .
-

Lemma 3.5. The output T of Algorithm 1 is a VEDs of the UDG corresponding to P and $|T| \leq 4|OPT|$, where OPT denotes a minimum size subset of P such that OPT vertex-edge dominates all the edges in the UDG corresponding to P .

Proof. Observe the way we choose T , $T = OPT_A \cup OPT_B \cup OPT_C \cup OPT_D$ and each OPT_i vertex-edge dominates all the edges in the combined-cells colored with i . As all the edges of $\mathcal{R}$ are a part of at least one combined-cell colored with a specific color i and vertex-edge dominated by at least one point of OPT_i , T is a VEDs of $\mathcal{R}$. The cardinality of T follows from the fact that each $|OPT_i| \leq |OPT|$ for all $i = A, B, C, D$. $\square$

Lemma 3.6. The running time of Algorithm 1 is polynomial.

Proof. Follows from the **for** loop in line number 4 of the algorithm and the execution time of line number 5, which is polynomial (see Lemma 3.7). $\square$

Computing an optimum solution for a single combined-cell

Let P_2 be a subset of P such that each point of P_2 can vertex-edge dominate at least one edge in S . The pseudocode for computing an optimum solution for a single combined-cell is described in Algorithm 2.

Algorithm 2 Vertex_Edge_Domination_Single_Combined-cell(P, S)

Require: A point set P and a combined-cell S

Ensure: $S_0 \subseteq P$, a minimum size vertex-edge dominating set for the cell S .

- 1: Let P_2 be the subset of P such that each point of P_2 can vertex-edge dominate at least one edge in S .
 - 2: Find a minimum size vertex-edge dominating set S_0 for the edges S using the points in P_2 by exhaustive search starting from size 1
 - 3: Return S_0 .
-

Lemma 3.7. The running time of Algorithm 2 is $O(n_s^{30})$, where $n_s = |P_2|$.

Proof. Let OPT_S be a subset of P_2 of minimum size such that the points in OPT_S can vertex-edge dominate all the edges in S . Now, from Claim 3.2 we know that a single point from a hexagonal cell ensures vertex-edge domination of all the edges in that cell and the combined-cell S is the combination of 30 hexagonal cells. So $|OPT_S| \leq 30$.

In line number 2 of the algorithm, we consider all possible combinations of points in P_2 of size $i = 1, 2, \dots, 29$ respectively. For each combination, we check whether all the edges in S are vertex-edge dominated or not. If for any i , there exists a subset of P_2 of size i that vertex-edge dominates all the edges in S , then that subset is reported and execution stops. If this fails for all $i = 1, 2, \dots, 29$, then the optimum solution of the combined-cell S consists of any one point from each cell of S . Thus the lemma. $\square$

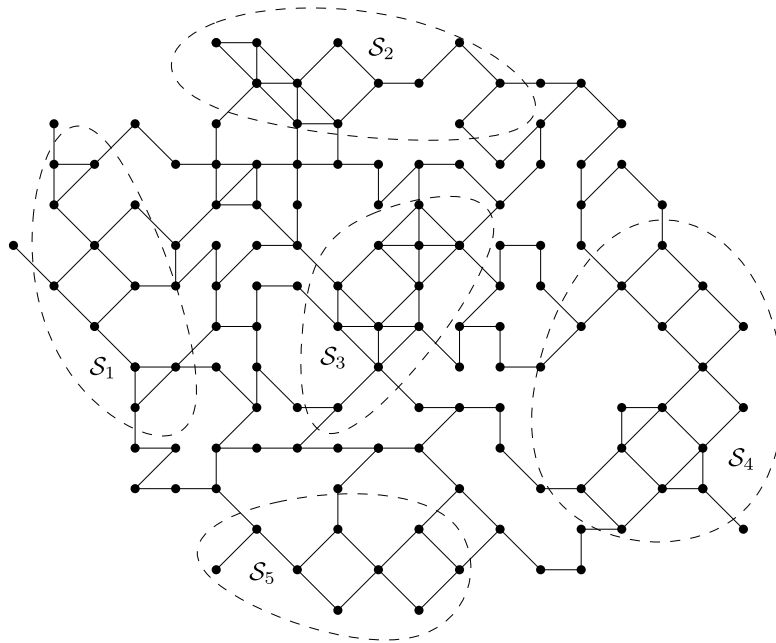


Fig. 6. A 4-separated collection $S = \{S_1, S_2, S_3, S_4, S_5\}$.

Theorem 3.8. The proposed 4-coloring scheme gives a 4-factor approximation algorithm for the VEDs problem in polynomial-time.

Proof. The approximation factor follows from Lemma 3.5. The time complexity result of the theorem follows from Lemma 3.6. $\square$

4. Approximation scheme

In this section, we propose a PTAS for the VEDs problem in UDGs. Let $G = (V, E)$ be a given UDG. The PTAS is based on the concept of m -separated collection of subsets of V for some integer m . Given a graph G , let $d(u, v)$ denote the number of edges on a shortest path between u and v . For $V_1, V_2 \subseteq V$, $d(V_1, V_2)$ is defined as $d(V_1, V_2) = \min_{u \in V_1, v \in V_2} \{d(u, v)\}$. We use notations $VED(A)$ and $VED_{opt}(A)$ to denote a vertex-edge dominating set of the graph induced by A ($\subseteq V$) in G and an optimal vertex-edge dominating set of the graph induced by A in G . We also define the closed neighborhood of a set $A \subseteq V$ as $N_G[A] = \bigcup_{v \in A} N_G[v]$ and the r th neighborhood of a vertex v as $N_G^r[v] = \{u \in V \mid d(u, v) \leq r\}$ in G .

Definition 4.1 (m -separated Collection). Let $S = \{S_1, S_2, \dots, S_k\}$ be a collection of disjoint vertex subsets in G such that each $S_i \subseteq V$ for $i = 1, 2, \dots, k$. S is referred as a m -separated collection of vertices if $d(S_i, S_j) > m$, for $1 \leq i, j \leq k$ and $i \neq j$ (see Fig. 6 for a 4-separated collection).

Nieberg and Hurink [19] considered 2-separated collection to propose a PTAS for the minimum dominating set problem in unit disk graphs.

Lemma 4.2. If $S = \{S_1, S_2, \dots, S_k\}$ is a m -separated collection in a graph $G = (V, E)$, then $\sum_{i=1}^k |VED_{opt}(S_i)| \leq |VED_{opt}(V)|$ for each $m \geq 4$.

Proof. For each $S_i \in S$, consider $P_i = \{u \in V \mid v \in S_i \text{ and } d(u, v) \leq 2\}$, for $i = 1, 2, \dots, k$. Since $m \geq 4$, $P_i \cap P_j = \emptyset$ as $d(S_i, S_j) > m$ for $i \neq j$. Observe that, for each $i = 1, 2, \dots, k$, $S_i \subseteq P_i$ and $P_i \cap VED_{opt}(V)$ is a vertex-edge dominating set of S_i . Therefore, $(P_i \cap VED_{opt}(V)) \cap (P_j \cap VED_{opt}(V)) = \emptyset$, and hence, we have $\sum_{i=1}^k |(P_i \cap VED_{opt}(V))| \leq |VED_{opt}(V)|$. As $P_i \cap VED_{opt}(V)$ is a vertex-edge dominating set of S_i , for $i = 1, 2, \dots, k$, and $VED_{opt}(V)$ is a minimum vertex-edge dominating set of the graph G , we obtain $\sum_{i=1}^k |VED_{opt}(S_i)| \leq \sum_{i=1}^k |(P_i \cap VED_{opt}(V))| \leq |VED_{opt}(V)|$. $\square$

Lemma 4.3. Let $S = \{S_1, S_2, \dots, S_k\}$ be an m -separated collection in a graph $G = (V, E)$, $m \geq 4$, and let $\mathcal{R}_1, \mathcal{R}_2, \dots, \mathcal{R}_k$ be subsets of V with $S_i \subseteq \mathcal{R}_i$ for all $i = 1, 2, \dots, k$. If there exists $\rho \geq 1$ such that $|VED_{opt}(\mathcal{R}_i)| \leq \rho |VED_{opt}(S_i)|$ holds for all

$i = 1, 2, \dots, k$, and if $\bigcup_{i=1}^k \text{VED}_{\text{opt}}(\mathcal{R}_i)$ is a vertex-edge dominating set in G , then $\sum_{i=1}^k |\text{VED}_{\text{opt}}(\mathcal{R}_i)|$ is at most ρ times the size of a minimum vertex-edge dominating set in G .

Proof. $\sum_{i=1}^k |\text{VED}_{\text{opt}}(\mathcal{S}_i)| \leq |\text{VED}_{\text{opt}}(V)|$ (from Lemma 4.2).

Hence, $\sum_{i=1}^k |\text{VED}_{\text{opt}}(\mathcal{R}_i)| \leq \rho \sum_{i=1}^k |\text{VED}_{\text{opt}}(\mathcal{S}_i)| \leq \rho |\text{VED}_{\text{opt}}(V)|$. $\square$

4.1. Construction of subsets

In this section, we discuss the process of constructing the desired 4-separated collection of subsets $\mathcal{S} = \{\mathcal{S}_1, \mathcal{S}_2, \dots, \mathcal{S}_k\}$ and the corresponding subsets $\mathcal{R}_1, \mathcal{R}_2, \dots, \mathcal{R}_k$ of V such that $\mathcal{S}_i \subseteq \mathcal{R}_i$ for all $i = 1, 2, \dots, k$. The main idea of constructing such subsets $\mathcal{S} = \{\mathcal{S}_1, \mathcal{S}_2, \dots, \mathcal{S}_k\}$ and the corresponding subsets $\mathcal{R}_1, \mathcal{R}_2, \dots, \mathcal{R}_k$ of V such that $\mathcal{S}_i \subseteq \mathcal{R}_i$ for all $i = 1, 2, \dots, k$ is to propose a PTAS for the VEDs problem with the help of Lemma 4.3. We first discuss the process of obtaining 4-separated collection of subsets $\mathcal{S} = \{\mathcal{S}_1, \mathcal{S}_2, \dots, \mathcal{S}_k\}$ and the corresponding subsets $\mathcal{R}_1, \mathcal{R}_2, \dots, \mathcal{R}_k$ of V such that $\mathcal{S}_i \subseteq \mathcal{R}_i$ for all $i = 1, 2, \dots, k$, then we prove that there exists a $\rho \geq 1$ such that $|\text{VED}_{\text{opt}}(\mathcal{R}_i)| \leq \rho |\text{VED}_{\text{opt}}(\mathcal{S}_i)|$ holds for all $i = 1, 2, \dots, k$, and finally we proved that $\bigcup_{i=1}^k \text{VED}_{\text{opt}}(\mathcal{R}_i)$ is a vertex-edge dominating set in G to obtain the PTAS for the problem.

The algorithm to find 4-separated collection of subsets in the graph G is discussed below: the algorithm proceeds in an iterative manner. It starts with an arbitrary vertex $v \in V_i$, where V_i is the vertex set in the i th iteration of the algorithm. Note that in the first iteration $V_1 = V$ and the algorithm computes $\mathcal{S}_1$ and $\mathcal{R}_1$. More specifically, for $r = 1, 2, \dots$, we find the vertex-edge dominating set of the graphs induced by the r th neighborhood as well as the $(r+4)$ -th neighborhood of the vertex v until $|\text{VED}(N_G^{r+4}[v])| > \rho |\text{VED}(N_G^r[v])|$ holds. Here, $\text{VED}(N_G^{r+4}[v])$ and $\text{VED}(N_G^r[v])$ are vertex-edge dominating sets of the graphs induced by $N_G^{r+4}[v]$ and $N_G^r[v]$, respectively, and $\rho = 1 + \epsilon$ ($\epsilon > 0$). Let $\hat{r}$ be the smallest r violating the above condition (see Lemma 4.6 for existence of such $\hat{r}$). Set $\mathcal{S}_i = N_G^{\hat{r}}[v]$, $\mathcal{R}_i = N_G^{\hat{r}+4}[v]$ and $V'_i = V_i \setminus N_G^{\hat{r}+3}[v]$. Note that removing $N_G^{\hat{r}+4}[v]$ from V_i implies removing the relevant edges connecting $N_G^{\hat{r}+4}[v]$ to $V_i \setminus N_G^{\hat{r}+4}[v]$ for which vertex-edge domination may not be maintained. Hence, removing $N_G^{\hat{r}+3}[v]$ from V_i removes the edges for which $\text{VED}(N_G^{\hat{r}+4}[v])$ is a vertex-edge dominating set. Let T_i be the set of vertices consisting of all singleton vertices after removing $N_G^{\hat{r}+3}[v]$ vertices from V_i in the i th iteration of the algorithm. Set $V_{i+1} = V'_i \setminus T_i$. The process stops while $V_{i+1} = \emptyset$ and returns the sets $\mathcal{S} = \{\mathcal{S}_1, \mathcal{S}_2, \dots, \mathcal{S}_k\}$ and $\mathcal{R} = \{\mathcal{R}_1, \mathcal{R}_2, \dots, \mathcal{R}_k\}$. The pseudocode for computing $\mathcal{S}$ and $\mathcal{R}$ is described in Algorithm 3. Note that the collection of the sets $\mathcal{S}$ is a 4-separated collection.

Algorithm 3 Vertex-edge_dominating_set (G)

Require: An undirected graph $G = (V, E)$ and an arbitrary small real number $\epsilon > 0$

Ensure: The sets $\mathcal{S} = \{\mathcal{S}_1, \mathcal{S}_2, \dots, \mathcal{S}_k\}$ and $\mathcal{R} = \{\mathcal{R}_1, \mathcal{R}_2, \dots, \mathcal{R}_k\}$

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1:  $i = 0, r = 1$  and  $V_{i+1} = V$ 
2:  $\mathcal{S} = \emptyset, \mathcal{R} = \emptyset$  and  $\rho = 1 + \epsilon$ 
3: while ( $V_{i+1} \neq \emptyset$ ) do
4:   pick an arbitrary  $v \in V_{i+1}$ 
5:    $N^0[v] = v$ 
6:    $r = 1$ 
7:   while  $|\text{VED}(N_G^{r+4}[v])| > \rho |\text{VED}(N_G^r[v])|$  do  $\triangleright \text{VED}(N_G^r[v]) = \text{MIS}(N_G^r[v])$ 
8:      $r = r + 1$ 
9:   end while
10:   $\hat{r} = r$   $\triangleright$  the smallest  $r$  violating while condition in step 7
11:   $i = i + 1$   $\triangleright$  the index  $i$  keeps track of the number of iterations
12:   $\mathcal{S}_i = N_G^{\hat{r}}[v]$ 
13:   $\mathcal{R}_i = N_G^{\hat{r}+4}[v]$ 
14:   $V'_i = V_i \setminus N_G^{\hat{r}+3}[v]$ 
15:   $T_i \leftarrow$  singleton vertices after removing  $N_G^{\hat{r}+3}[v]$ 
16:   $V_{i+1} = V'_i \setminus T_i$ 
17: end while
18: Return  $\mathcal{S} = \{\mathcal{S}_1, \mathcal{S}_2, \dots, \mathcal{S}_k\}$  and  $\mathcal{R} = \{\mathcal{R}_1, \mathcal{R}_2, \dots, \mathcal{R}_k\}$ .

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Let $\text{VED}(N_G^r[v])$ be an arbitrary maximal independent set of the subgraph $G' = (V', E')$ induced by the vertices in $N_G^r[v]$. Next, we show that (i) $\text{VED}(N_G^r[v])$ is a vertex-edge dominating set of G' (see Lemma 4.4), and (ii) the size of the $\text{VED}(N_G^r[v])$ is bounded by $(r+2)^2$ (see Lemma 4.5).

Lemma 4.4. $\text{VED}(N_G^r[v])$, a maximal independent set of the graph G' induced by the vertices in $N_G^r[v]$, is a VEDs of $N_G^r[v]$ in G .

Proof. On the contrary, assume that $VED(N_G^r[v])$ is not a VEDs of the graph $G' = (V', E')$ induced by the vertices in $N_G^r[v]$. That means, there exists an edge $xy \in E'$ such that $N_{G'}[x] \notin VED(N_G^r[v])$ and $N_{G'}[y] \notin VED(N_G^r[v])$. It contradicts the fact that $I = VED(N_G^r[v])$ is a maximal independent set in G' . Thus, the lemma. $\square$

Lemma 4.5. $|VED(N_G^r[v])| \leq (r+2)^2$.

Proof. The cardinality of a maximal independent set in the UDG G' , a subgraph induced by the vertices in $N_G^r[v]$, is bounded by the number of non-intersecting unit disks packed in a disk of radius $r+2$ centered at v . Since $VED(N_G^r[v])$ is also an independent set, $|VED(N_G^r[v])| \leq \frac{\pi(r+2)^2}{\pi(1)^2} = (r+2)^2$. Therefore, $|VED(N_G^r[v])| \leq (r+2)^2$. $\square$

Lemma 4.6. For $\rho = 1 + \epsilon$, there always exists an r violating the condition $|VED(N_G^{r+4}[v])| > \rho |VED(N_G^r[v])|$.

Proof. On the contrary, assume that there exists a vertex $v \in V$ such that

$|VED(N_G^{r+4}[v])| > \rho |VED(N_G^r[v])|$ for all $r = 1, 2, \dots$

From Lemma 4.5, we have $|VED(N_G^r[v])| \leq (r+2)^2$.

Therefore, if $r = 4k$,

$(r+2)^2 \geq |VED(N_G^r[v])| > \rho |VED(N_G^{r-4}[v])| > \dots > \rho^{\frac{r}{4}} |VED(N_G^2[v])| \geq \rho^{\frac{r}{4}}$, and if $r = 4k + s$ for $1 \leq s \leq 3$,

$(r+2)^2 \geq |VED(N_G^r[v])| > \rho |VED(N_G^{r-4}[v])| > \dots > \rho^{\frac{r-1}{4}} |VED(N_G^1[v])| \geq \rho^{\frac{r-1}{4}}$.

Hence,

$$(r+2)^8 > \begin{cases} \rho^r, & \text{if } r \text{ is } 4k \\ \rho^{r-1}, & \text{if } r \text{ is } 4k + s \end{cases} \quad (1)$$

Observe that in the inequality (1), the right side is an exponential function where as the left side is a polynomial function in r , which results in a contradiction. $\square$

Lemma 4.7. The smallest r violating inequality (1) is bounded by $O(\frac{1}{\epsilon} \log \frac{1}{\epsilon})$.

Proof. Let $\hat{r}$ be the smallest r violating the inequalities (1). We prove $\hat{r} \leq O(\frac{1}{\epsilon} \log \frac{1}{\epsilon})$ by using the inequalities (i) $\log(1 + \epsilon) > \frac{\epsilon}{2}$ for $0 < \epsilon < 1$, (ii) $\log x < x$ for $x > 1$, and (iii) $\log \frac{1}{\epsilon} \geq 1$ for $\epsilon \leq \frac{1}{10}$. Let $x = \frac{\epsilon}{\log \frac{1}{\epsilon}}$. Consider the inequality $(x+2)^8 \leq (2x)^8 \leq (1+\epsilon)^x$. The former inequality trivially holds for the choice of x and for any $\epsilon > 0$, and taking the logarithm on both sides of the latter inequality, we get $\frac{8 \log 2x}{x} \leq \log(1 + \epsilon)$. By inequality (i), now, it suffice to show that $\frac{8 \log 2x}{x} \leq \frac{\epsilon}{2}$. Using the inequalities (ii) and (iii), the choice of x satisfies the inequality for any constant c satisfying $\log 2c < \frac{c}{16}$. $\square$

Lemma 4.8. For a given $v \in V$, a minimum vertex-edge dominating set $VED_{\text{opt}}(\mathcal{R}_i)$ of $\mathcal{R}_i$ can be computed in polynomial time.

Proof. Let $G' = (V', E')$ be a graph induced by $\mathcal{R}_i \subseteq N_G^{r+4}[v]$. From Lemma 4.5, the size of $N_G^{r+4}[v]$ is bounded by $O(r^2)$, so we take every possible tuple of size at most $O(r^2)$ and check whether the selected tuple is a vertex-edge dominating set of the graph G' . This process takes $O(\binom{n}{r^2}) = O(n^{r^2})$ time. Since $r = O(\frac{1}{\epsilon} \log \frac{1}{\epsilon})$ by Lemma 4.7, $VED_{\text{opt}}(\mathcal{R}_i)$ can be computed in polynomial-time. $\square$

Lemma 4.9. For the collection of subsets $\{\mathcal{R}_1, \mathcal{R}_2, \dots, \mathcal{R}_k\}$, $\mathcal{D} = \bigcup_{i=1}^k VED(\mathcal{R}_i)$ is a vertex-edge dominating set in $G = (V, E)$.

Proof. To prove $\mathcal{D}$ is a vertex-edge dominating set of the graph G , we need to prove for every edge $v_i v_j \in E$, there exists at least one vertex from $N_G[v_i]$ or $N_G[v_j]$ in $\mathcal{D}$. It follows from our construction of the subsets $\mathcal{R}_1, \mathcal{R}_2, \dots, \mathcal{R}_k$ (Section 4.1) that each edge $v_i v_j$ belongs to a particular subset $\mathcal{R}_i$ and $VED(\mathcal{R}_i)$ is a vertex-edge dominating set of the graph induced by the vertices of $\mathcal{R}_i$. Thus the lemma. $\square$

Corollary 4.10. $\mathcal{D}^* = \bigcup_{i=1}^k VED_{\text{opt}}(\mathcal{R}_i)$ is a vertex-edge dominating set in G , for the collection $\mathcal{R} = \{\mathcal{R}_1, \mathcal{R}_2, \dots, \mathcal{R}_k\}$.

Theorem 4.11. For a given UDG, $G = (V, E)$, and an $\epsilon > 0$, we can design a $(1 + \epsilon)$ -factor approximation algorithm to find a VEDs in G with running time $n^{O(c^2)}$, where $c = O(\frac{1}{\epsilon} \log \frac{1}{\epsilon})$.

Proof. The proof of the theorem follows from Lemmas 4.3, 4.6, 4.8 and Corollary 4.10. $\square$

5. Conclusion and future directions

In this article, we studied the minimum vertex-edge domination problem in unit disk graphs and showed that the decision version of the problem is NP-complete. Next, we proposed a simple polynomial-time 4-factor approximation

algorithm for the problem. We have also shown that the vertex-edge domination problem in unit disk graphs admits a PTAS.

It remains open whether there exists a better constant-factor approximation algorithm for the problem in UDGs. In particular, an $O(n^c)$ -time constant-factor algorithm, where $c \ll 30$ would provide a natural polynomial-time constant-factor approximation algorithm. It would be interesting to see how the problem behaves in different special subclasses of graphs.

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